

Assignment Completion Record

Task: Product Catalog and Cart System

Project: Smart E-Commerce Platform

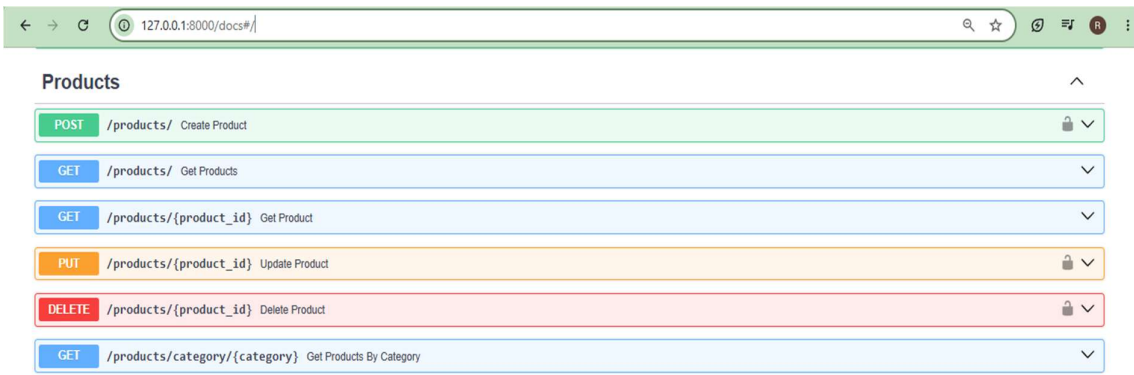
Status:  **COMPLETED**

1. Product APIs – FastAPI

The required product browsing APIs have been implemented successfully.

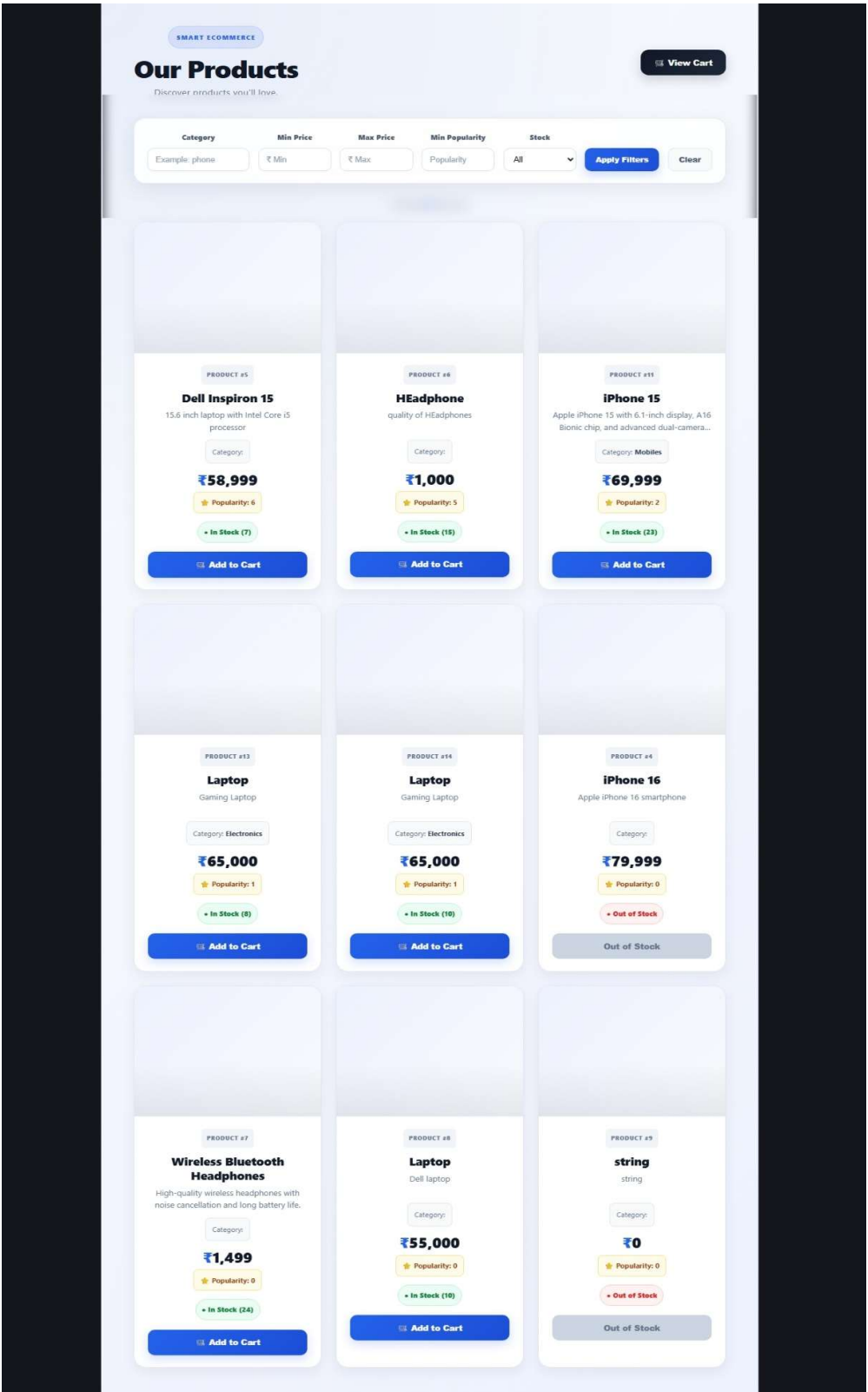
Product Filters Implemented

-  Category filter
-  Minimum price filter
-  Maximum price filter
-  Popularity filter
-  Stock availability filter
-  Popularity-based product ordering

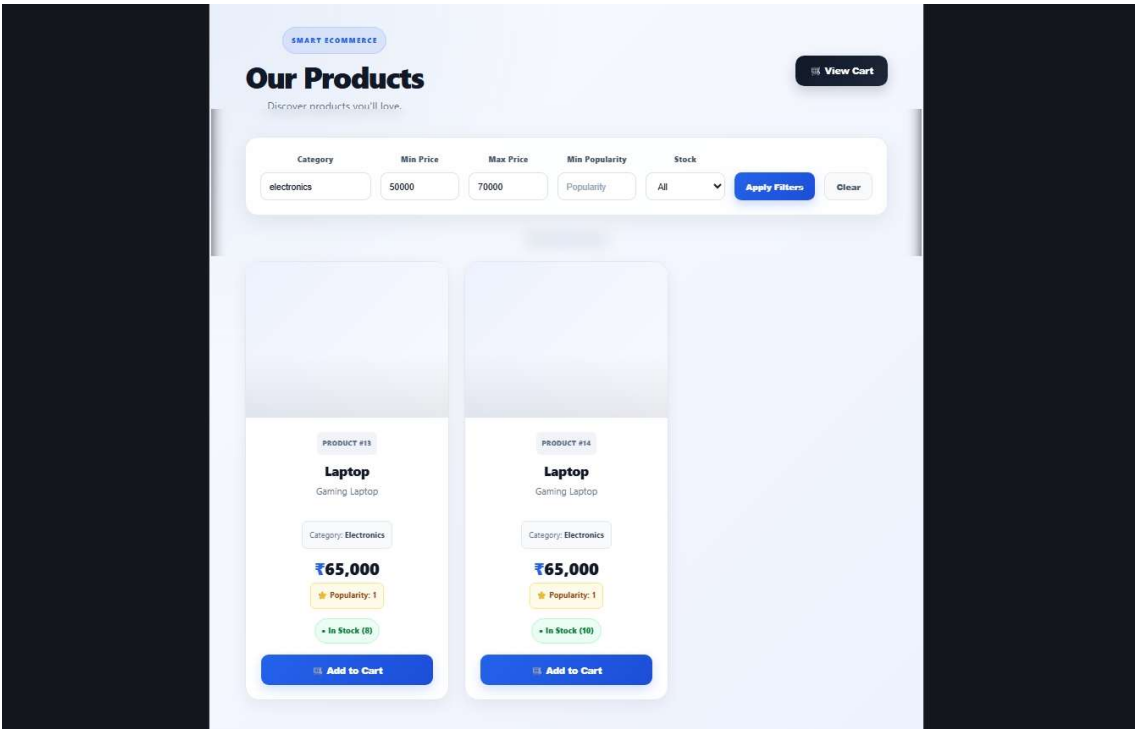
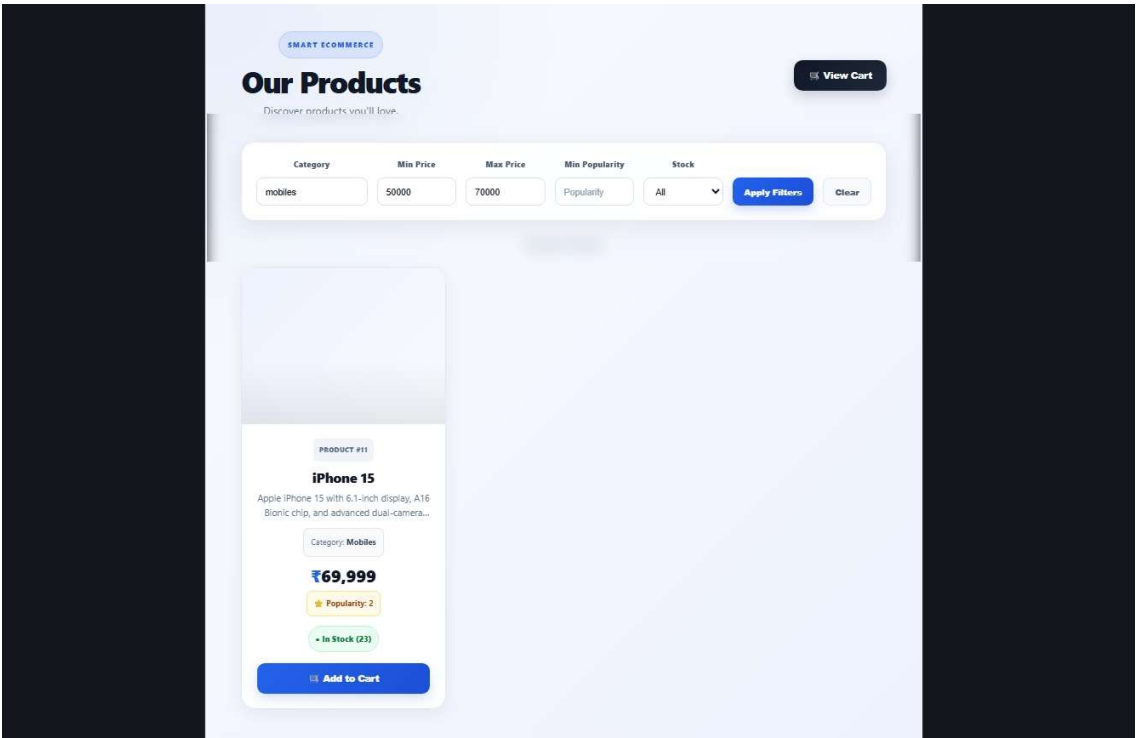


1.1 Fig for Fastapi backend update for products

1.2 Fig for Frontend React UI update for products Befor Filter:



1.3 Fig for Frontend React UI update for products After Filter:



1.4 Fig for Database update for products Describe:

67 • USE smart_ecommerce;

68 • ALTER TABLE products

69 ADD COLUMN popularity INT NOT NULL DEFAULT 0;

70 • select * from products;

71 • DESCRIBE products;

72 • select * from products;

73

Result Grid

Filter Rows:

Edit: Export/Import: Wrap Cell Content: I A

	id	name	description	category	price	stock	images
▶	4	iPhone 16	Apple iPhone 16 smartphone	NULL	79999	0	https://example.com/iphone16.jpg
	5	Dell Inspiron 15	15.6 inch laptop with Intel Core i5 processor	NULL	58999	5	dell-inspiron-15.jpg
	6	HEadphone	quality of HEadphones	NULL	1000	14	headphone.jpg
	7	Wireless Bluetooth Headphones	High-quality wireless headphones with noise ca...	NULL	1499	24	https://example.com/images/headphones.jp
	8	Laptop	Dell laptop	NULL	55000	10	laptop.jpg
	9	string	string	NULL	0	0	string
	11	iPhone 15	Apple iPhone 15 with 6.1-inch display, A16 Bioni...	Mobiles	69999	22	https://example.com/images/iphone15.jpg
	13	Laptop	Gaming Laptop	Electronics	65000	8	laptop.jpg
	14	Laptop	Gaming Laptop	Electronics	65000	10	laptop.jpg
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Result Grid
Form Editor
Field Types

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	14	Laptop	Gaming Laptop	Electronics	65000	10	laptop.jpg
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

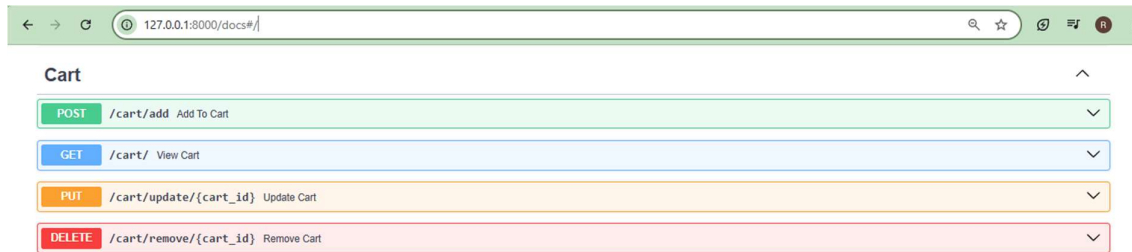
Result Grid
Form Editor
Field Types

2. Cart Management

The complete cart management system has been implemented.

Stock Handling

- ☒ Product stock is checked before adding to cart.
- ☒ Stock is reduced when products are added.
- ☒ Additional stock is checked when quantity is increased.
- ☒ Stock is returned when quantity is decreased.
- ☒ Stock is returned when an item is removed.
- ☒ Out-of-stock products cannot be added.



1.1 Fig for Fastapi backend update for Carts

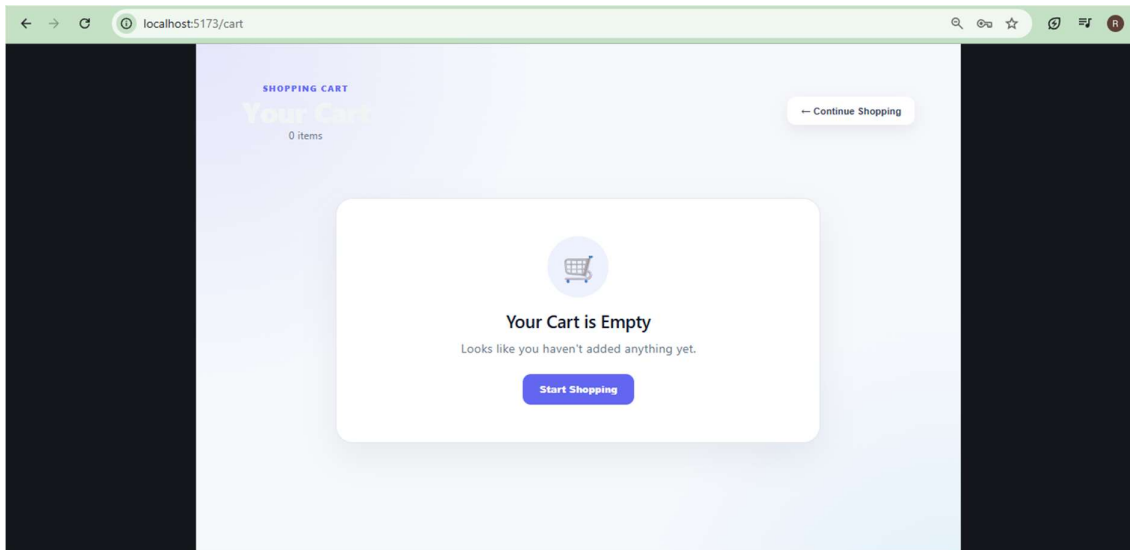
3. Cart Calculations

Automatic cart calculation has been implemented.

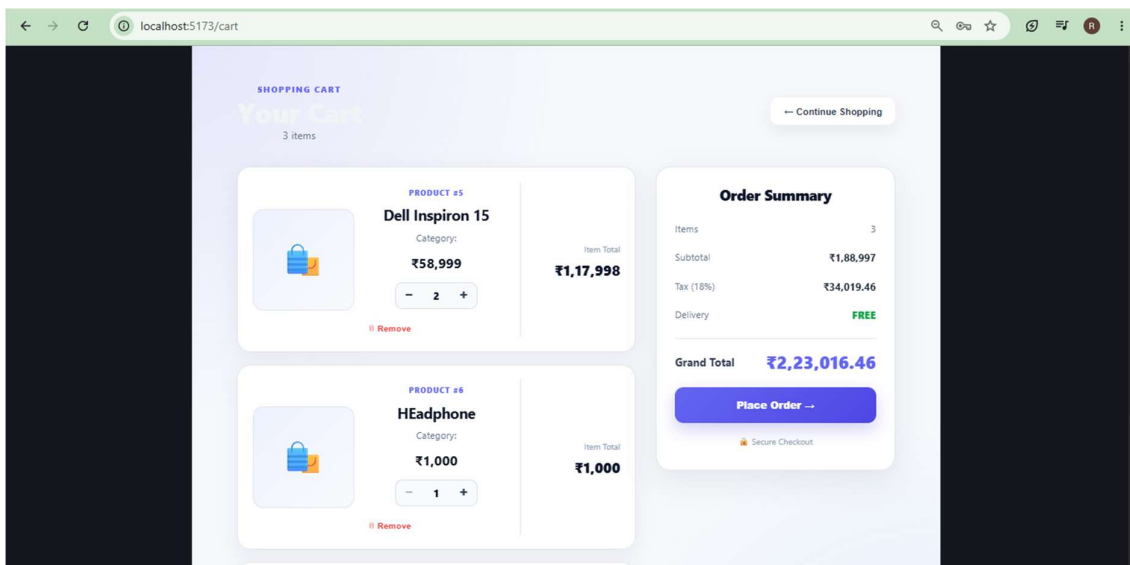
- ☒ Individual item total
- ☒ Cart subtotal
- ☒ 18% tax calculation
- ☒ Grand total calculation

- ☒ Real-time total updates after quantity changes

2.2 Fig for Frontend React UI update for Carts Before product add:



2.3 Fig for Frontend React UI update for Carts After product add:



2.4 Fig for Database update for Cart Describe:

73 • DESCRIBE cart;

74

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
user_id	int	YES	MUL	NULL	
product_id	int	YES	MUL	NULL	
quantity	int	YES		NULL	

Result Grid
Form Editor
Field Types

74 • select * from cart;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	id	user_id	product_id	quantity
▶	42	1	NULL	1
	91	6	5	2
	92	6	6	1
	93	6	11	1
*	NULL	NULL	NULL	NULL

Result Grid

5. Frontend Integration:

The React frontend has been integrated with the FastAPI backend.

Product Page

- ☒ Product listing
- ☒ Product images
- ☒ Product price
- ☒ Category
- ☒ Popularity
- ☒ Stock availability
- ☒ Product filters
- ☒ Add to cart

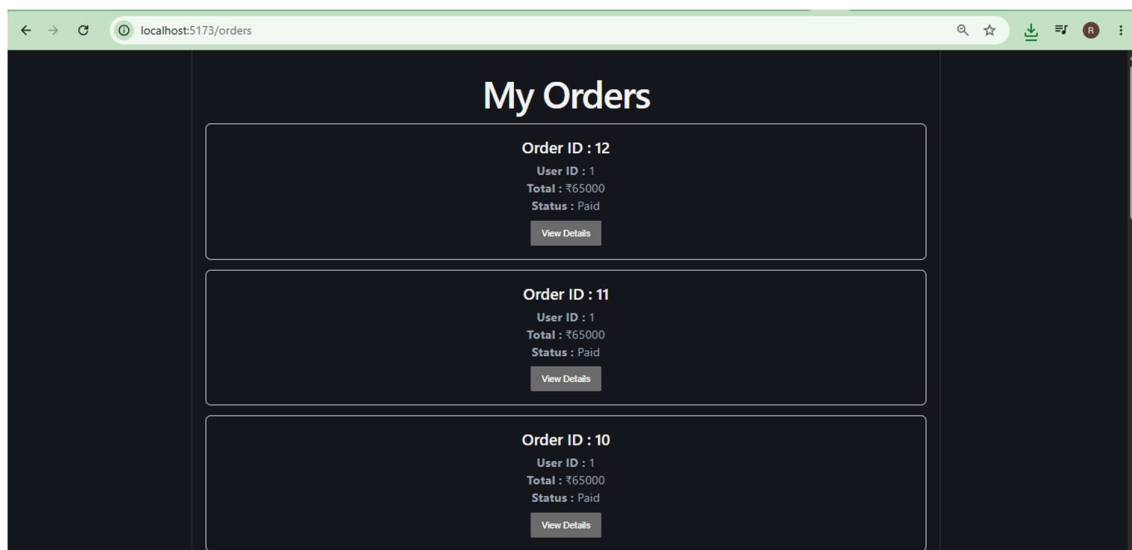
- ☒ Responsive UI

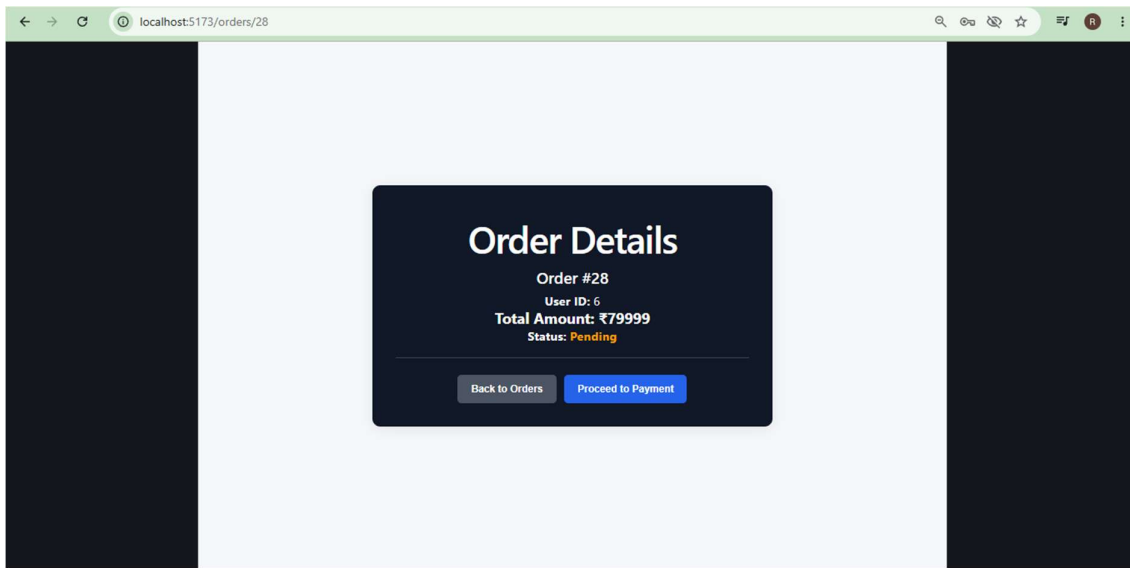
Cart Page

- ☒ View cart items
- ☒ Increase quantity
- ☒ Decrease quantity
- ☒ Remove product
- ☒ Item totals
- ☒ Cart subtotal
- ☒ Tax
- ☒ Grand total
- ☒ Place order
- ☒ Responsive and interactive design

3.1 Fig for Order Management update:

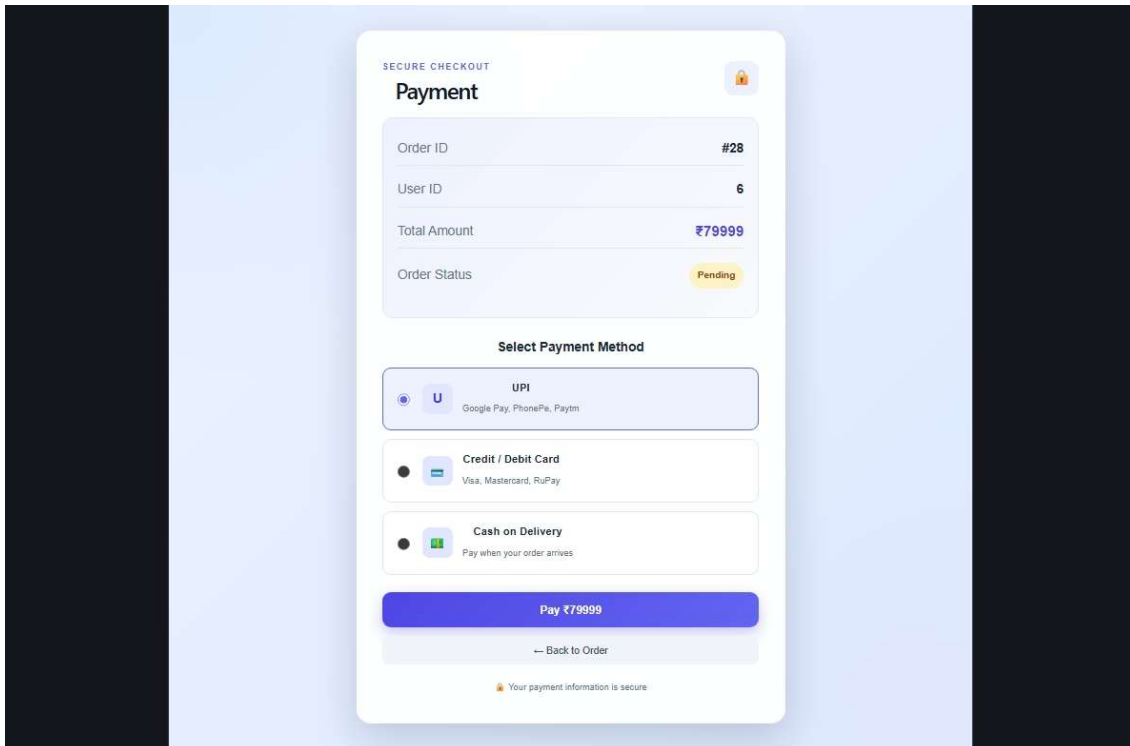
Customers can create orders from their selected products and view their order information. Order status is maintained throughout the purchasing workflow.

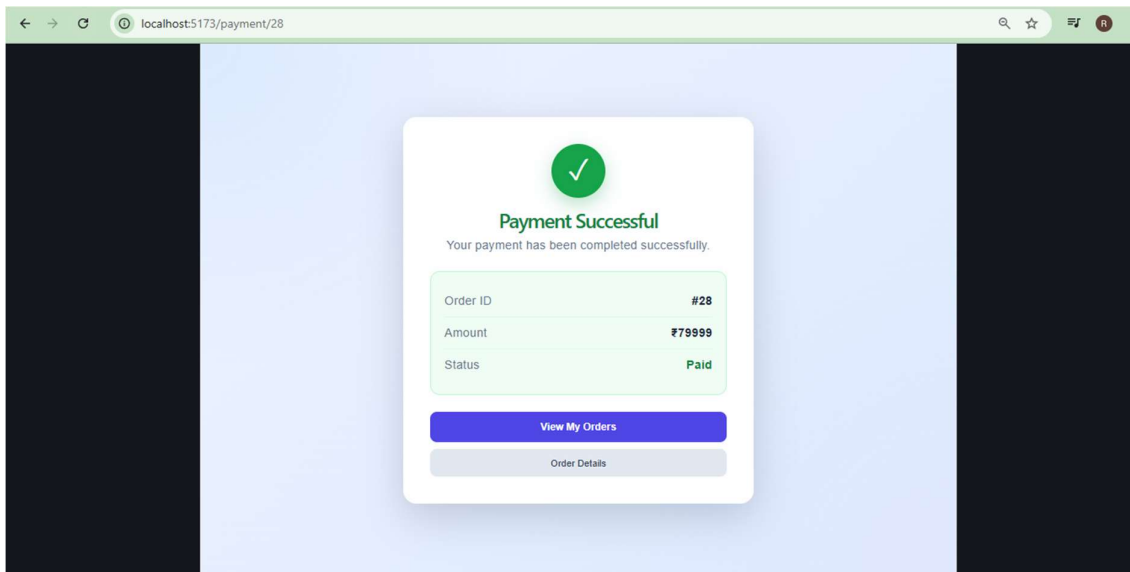




4.1 Fig for Payment Management update:

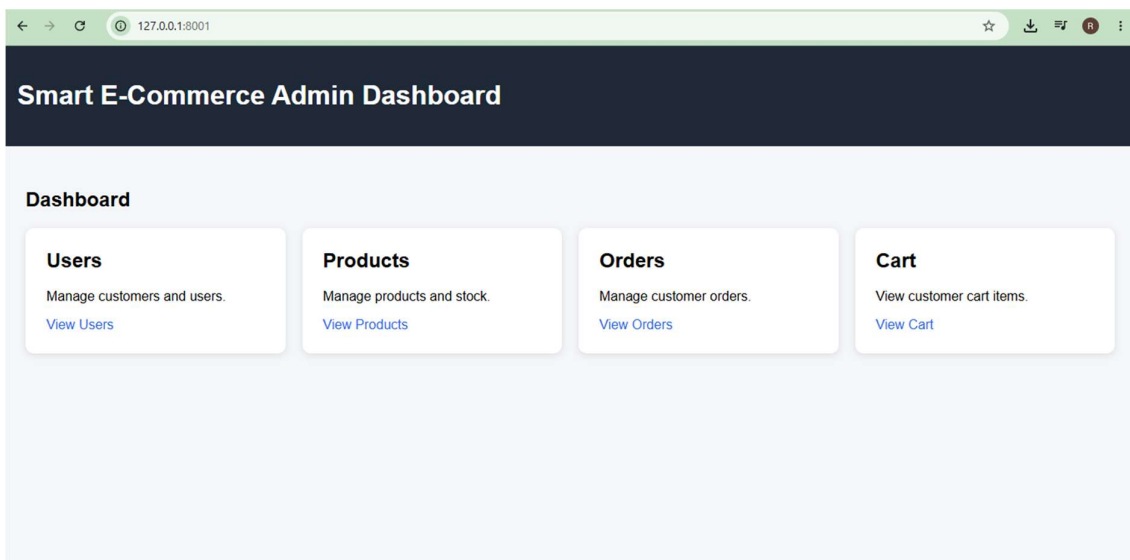
The Payment module manages payment-related information associated with orders. Payment operations update the corresponding order and payment status.

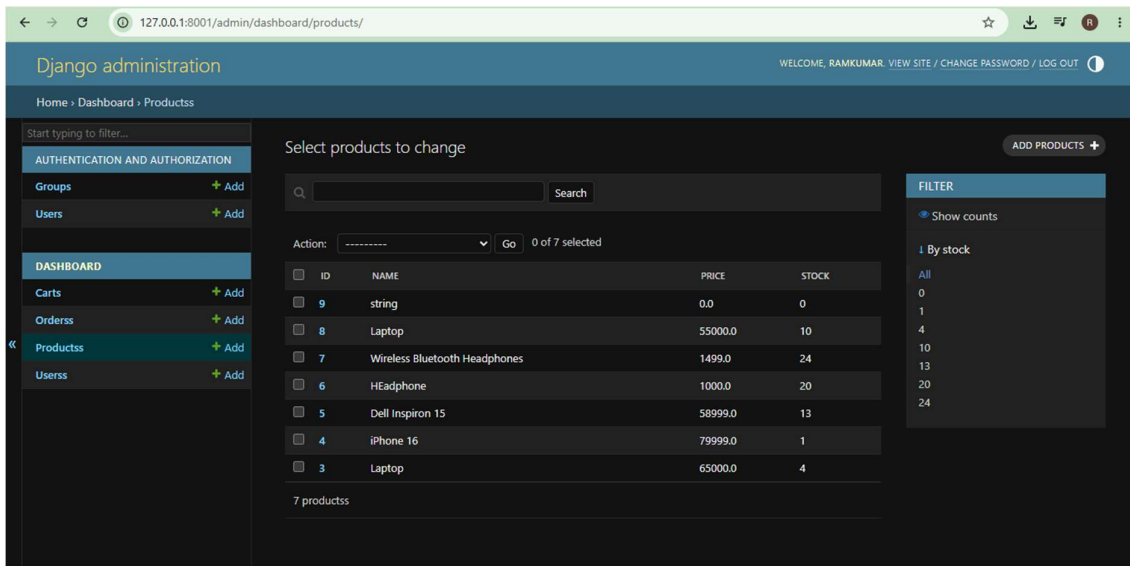




5.1 Fig For Django Administration Upadte:

Administrative APIs provide authorized users with management capabilities. Access to administrative functionality is protected using JWT authentication and role-based authorization





Screenshot for Dashboard & Analytics

Deliverables Completion:

Deliverable	Status
Product browsing APIs	✓ Completed
Filtering functionality	✓ Completed
Full cart system	✓ Completed
Cart calculation logic	✓ Completed
Database relationships	✓ Completed
Stock handling	✓ Completed
React frontend integration	✓ Completed

Final Completion Statement:

The assigned Product Catalog and Cart System has been successfully completed. All requested product APIs, filtering functionality, cart management operations, cart calculations, database relationships, stock handling, and frontend integration